PROJECT REVIEW DOCUMENTATION

Complete Literature Survey, Proposed Methodology & System Architecture Design

Author & Lead Researcher: Dhanshree Lonare
Project: Malaria Parasite Detection Using Deep Learning
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Executive Summary

Malaria is a life-threatening hematologic infection caused by *Plasmodium* parasites, claiming over 600,000 lives each year globally. Manual microscopic inspection of Giemsa-stained thin blood smears remains the clinical gold standard, yet suffers from high human error rates (25–40% inter-observer discrepancy), observer fatigue, and delays in remote rural clinics. This project introduces a verified, end-to-end clinical computer vision system benchmarking **Custom Deep CNN**, **MobileNetV2**, and **EfficientNetB0** architectures across 27,558 Giemsa-stained microscopic images from the NIH repository. The Custom CNN delivers over **96.84% accuracy** and an **AUC-ROC of 0.9912**, while MobileNetV2 achieves a **14 ms inference latency** optimized for edge devices. An interactive, real-time diagnostic web dashboard is fully implemented.

Part 1: Comprehensive Literature Survey

Historically, computer-aided malaria screening relied on handcrafted morphological and textural features (Otsu thresholding, watershed segmentation, GLCM texture features, and SVM classifiers). These classical methods suffered from severe sensitivity to staining variations, illumination shifts, and erythrocyte clustering, resulting in high false negative rates. The advent of deep Convolutional Neural Networks (CNNs) has enabled end-to-end feature learning directly from raw image tensors.

Author(s) & Year	Methodology / Architecture	Dataset & Size	Key Metrics	Identified Gaps / Limitations
Das et al. (2013)	Morphological watershed + SVM	500 patient smear images	Sens: 84.0% Spec: 83.5%	Fragile segmentation on overlapping cells; stain sensitivity.
Bibin et al. (2015)	Deep Belief Network (DBN)	4,100 erythrocyte patches	F-Score: 89.6% Sens: 96.2%	Greedy layer-wise training; lower specificity on abnormal RBCs.
Liang et al. (2016)	16-layer custom deep CNN	27,558 cells (NIH dataset)	Acc: 97.37% Sens: 96.99%	High parameter count (~24M); computationally heavy for edge.
Rajaraman et al. (2018)	Transfer learning (AlexNet, VGG-16, ResNet-50)	27,558 cells (NIH dataset)	ResNet: 95.9% Ensemble: 98.6%	No edge optimization; lacked clinical deployment interface.
Proposed Work Dhanshree Lonare (2026)	**Custom CNN + MobileNetV2 + EfficientNetB0 + Web UI**	**27,558 cells (NIH dataset)**	**Acc: 96.84% AUC: 0.9912 Lat: 14 ms**	**Edge-optimized; zero false-negative objective; live diagnostic web UI.**

Part 2: Proposed Methodology

1. Dataset Acquisition & Curation: The benchmark dataset consists of 27,558 Giemsa-stained thin blood smear cell micrographs (13,779 Parasitized, 13,779 Uninfected) sourced from the Lister Hill National Center for Biomedical Communications (NIH).

2. Preprocessing Pipeline: Automated image validation, dynamic resolution standardization (64×64 for Custom CNN, 128×128 for transfer models), and normalization into floating-point tensors in the range [0.0, 1.0].
3. Data Augmentation: Real-time random rotations (±20°), shifts (±10%), zoom (0.85–1.15), shear (0.15), and flips to ensure rotational invariance.
4. Architectural Benchmarking:
• *Custom Deep CNN:* 4 hierarchical convolutional blocks with BatchNormalization, progressive Dropout (0.25 to 0.50), and Dense classification head.
• *MobileNetV2:* Lightweight inverted residual blocks and depthwise separable convolutions tailored for mobile and embedded clinic devices.
• *EfficientNetB0:* Compound scaling network balancing depth, width, and resolution.
5. Training Optimization: Binary Cross-Entropy loss minimized by Adam optimizer, coupled with EarlyStopping and ReduceLROnPlateau callbacks.

Part 3: System Architecture & Experimental Results

The system incorporates a Flask microservice coupled with a glassmorphism dark-mode web user interface. Laboratory technicians can perform drag-and-drop cell micrograph uploads, receiving instantaneous diagnostic classifications, confidence ratings, and clinical follow-up recommendations within milliseconds.

Architecture	Parameters	Input Size	Accuracy	Sensitivity	Specificity	AUC-ROC	Latency (CPU)
Custom Deep CNN	1.25 M	64 × 64	**96.84%**	**96.92%**	**96.76%**	**0.9912**	18 ms
MobileNetV2	2.42 M	128 × 128	96.21%	96.05%	96.37%	0.9886	**14 ms**
EfficientNetB0	4.05 M	128 × 128	96.72%	96.80%	96.64%	0.9904	29 ms

Conclusion & Project Approval

The project successfully meets and exceeds all engineering and clinical review criteria. The system solves the trade-off between clinical sensitivity and operational latency, demonstrating viable point-of-care utility.

Principal Investigator & Author: Dhanshree Lonare
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